



HydroSovereign AI Engine

QGIS Plugin v6.01

Complete Professional User Manual — v5.0 · 12 Chapters · 4 Appendices

✓ OFFICIALLY PUBLISHED — QGIS Plugin Repository · Plugin ID: 5040 · April 21, 2026

Field	Details
Developer	Seifeldin M.G. Alkedir · University of Khartoum
ORCID	0000-0003-0821-2991
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Official URL	https://plugins.qgis.org/plugins/hsae_qgis/
QGIS Requirement	≥ 3.16 LTR (3.28 LTR recommended)
Security Scan	✓ Passed · 0 Critical · 0 Warnings · 76 files scanned
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PyPI	<code>pip install hydrosovereign (v6.5.3)</code>
Live App	https://hydrosovereign-ai-engine-hsae-v601-6euz2zxcmerkzvgordmvxf.streamlit.app
License	GPL-3.0 (GNU General Public License v3.0)
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Document Revision History

Version	Date	Changes
1.0	March 2026	Initial draft — 10 chapters
2.0	April 16, 2026	Added 13 tools, 5 algorithms, PR #289
3.0	April 21, 2026	Official publication ID 5040; security scan results; 21+ downloads
4.0	April 22, 2026	Added Chapter 11 — QGIS Desktop Step-by-Step Workflow (13 tools, 5 algorithms, complete workflow table, output folder tree, keyboard shortcuts, Print Layout, common errors)
5.0	April 24, 2026	Complete professional edition: Quick Reference Card (Appendix C); Workflow Diagram (Appendix D); Case Study 2 — Mekong Basin; Model Builder Guide (Ch.12); Batch Processing (Ch.12); Python API Integration (Ch.12); Data Preparation guide; annotated UI descriptions; full keyword Index (Appendix E); updated TOC; all internal cross-references verified

Table of Contents

Section	Page
Quick Start — 5-Minute Tutorial	3
Ch. 1 — Overview & Introduction	4
Ch. 2 — Installation & Requirements	5
2.1 Method A — QGIS Plugin Repository (Recommended)	5
2.2 Method B — Install from ZIP	5
2.3 Verifying the Installation	5
Ch. 3 — Interface Overview	6
3.1 HSAE Toolbar (11 buttons)	6
3.2 Menu-Only Tools	6
3.3 Processing Toolbox (5 Algorithms)	6
Ch. 4 — Complete Tool Reference (13 Tools)	7
Ch. 5 — Processing Toolbox Algorithms (5)	13
Ch. 6 — The 26 Registered Basins	16
Ch. 7 — Scientific Indices Reference	17
Ch. 8 — Google Earth Engine Data Sources	18
Ch. 9 — Troubleshooting & FAQ	19
Ch. 10 — Citation & References	20
Ch. 11 — QGIS Desktop Step-by-Step Workflow	21
11.1 Pre-Flight Checklist	21
11.2 Launching QGIS & Activating Plugin	21
11.3 Complete 13-Tool Workflow	22
11.4 Processing Algorithms Step-by-Step	30
11.5 Complete Workflow Summary (Blue Nile)	32
11.6 Expected Output Files	33
11.7 Keyboard Shortcuts	33
11.8 Working with Results	34
11.9 Common Desktop Errors & Solutions	34
Ch. 12 — Advanced Features	35
12.1 Model Builder Integration	35
12.2 Batch Processing (All 26 Basins)	36
12.3 Adding a New Basin	37
12.4 Python API Integration (hydrosovereign)	38
Ch. 13 — Case Study 2: Mekong River Basin	40
Appendix A — Glossary of Terms	43
Appendix B — All Links & Resources	44

Appendix C — Quick Reference Card (Print-Ready)	45
Appendix D — HSAE Workflow Diagram	46
Appendix E — Keyword Index	47

Quick Start — 5-Minute Tutorial

Follow these 5 steps to go from installation to your first basin analysis:

Step 1 — Install (1 minute)

1. Open QGIS → Plugins → Manage and Install Plugins → search 'HydroSovereign' → Install Plugin
2. **Or go directly:** https://plugins.qgis.org/plugins/hsae_qgis/
3.  HSAE toolbar appears immediately — no restart needed

Step 2 — Load Basins (10 sec)

4. Click  in the HSAE toolbar → 26 basin points appear on the world map

Step 3 — Apply Risk Colours (5 sec)

5. Click  (first) → basins recolour: Green=safe · Orange=Art.7 · Red=dispute

Step 4 — Open Dashboard (30 sec)

6. Click  → select 'Blue Nile (GERD)' → review: ATDI=43.5%, HIFD=20.0%, CI=0.44 HIGH

Step 5 — Export ICJ Dossier (30 sec)

7. Menu →  ICJ/PCA Dossier → save HTML → opens in browser with full legal assessment

 Blue Nile (GERD): ATDI=43.5% (Art.7 UNWC) · HIFD=20.0% · NSE=0.63 · CI=0.44 HIGH · P(Negotiation)=58%

Chapter 1 — Overview & Introduction

1.1 What is the HSAE QGIS Plugin?

The HydroSovereign AI Engine QGIS Plugin v6.01 (Plugin ID: 5040) is officially published in the QGIS Plugin Repository. It brings AI-powered transboundary water sovereignty analysis directly into QGIS Desktop, providing 13 interactive tools and 5 Processing Toolbox algorithms covering 26 globally contested river basins.

✓ Security scan: 0 Critical · 0 Warnings · 76 files scanned · Approved by zimbogisgeek (QGIS reviewer) · PR #289 — Tim Sutton, QGIS PSC

1.2 Does It Require a Separate QGIS Profile?

No. Installs into any existing QGIS profile (version 3.16+). No separate profile, no extra credentials required.

1.3 Plugin Components

Component	Count	Access
Toolbar Tools	13	HydroSovereign AI Engine v6.01 menu + toolbar
Processing Algorithms	5	Processing Toolbox → HydroSovereign AI Engine v6.01
Dashboard Panel	1 dockable	Tool 10 toggles open/close
Basin Registry	26 basins	Built-in basins_50.json — 7 world regions
GEE Satellite Scripts	7 sensors	Tool 5 generates GEE JavaScript
GRDC Stations	10 stations	Tool 4 loads as point layer

Chapter 2 — Installation & Requirements

2.1 System Requirements

Component	Minimum	Recommended
QGIS	3.16 LTR	3.28 LTR or newer
Python	3.8 (bundled)	3.10+ (bundled with QGIS 3.28)
OS	Windows 10, macOS 10.15, Ubuntu 20.04	Windows 11, macOS 13, Ubuntu 22.04
Internet	Required for GEE scripts	Stable broadband
Extra packages	None — all bundled	—

2.2 Method A — QGIS Plugin Repository (Recommended)

Plugin ID: 5040 · https://plugins.qgis.org/plugins/hsae_qgis/

8. Open QGIS
9. Plugins → Manage and Install Plugins
10. Search: HydroSovereign
11. Select 'HydroSovereign AI Engine' → Install Plugin
12.  Toolbar and menu appear immediately

 Security: 4/5 checks passed · 0 Critical · 0 Warnings · 76 files scanned (April 21, 2026)

2.3 Method B — Install from ZIP

13. Download ZIP from GitHub repository
14. Plugins → Manage → 'Install from ZIP' tab → browse to ZIP → Install

 Internal folder must be `hsae_qgis/` — do not rename. Method A is always preferred.

2.4 Verifying the Installation

Location	What You Should See
QGIS Menu Bar	'HydroSovereign AI Engine v6.01' between Plugins and Help
QGIS Toolbar	HSAE toolbar row with 11 emoji buttons
Processing Toolbox	'HydroSovereign AI Engine v6.01' group with 5 algorithms

2.5 Uninstalling

Plugins → Installed → Select 'HydroSovereign AI Engine' → Uninstall Plugin. All components removed cleanly.

Chapter 3 — Interface Overview

3.1 HSAE Toolbar — 11 Buttons

The HSAE toolbar appears as a horizontal row of emoji buttons in the QGIS toolbar area. From left to right:

P o s	Butt on	Tool Name	Keyboard
1		Load Basin Registry	—
2		TDI/ATDI Visualiser	—
3		UNWC Legal Layer	—
4		GRDC Stations Overlay	—
5		GEE Script Generator	—
6		Dashboard Dialog	—
7		Conflict Index	—
8		Negotiation AI	—
9		WebGIS Map Generator	—
10		Export Basin Data	—
11		Basin Panel (dockable)	—

3.2 Menu-Only Tools (require menu access)

Tool	Menu Path	Purpose
12 —  ICJ/PCA Dossier	Menu → HydroSovereign →  ICJ/PCA Dossier	Full legal dossier for all 26 basins
13 —  About	Menu → HydroSovereign →  About HSAE v6.01	Plugin metadata, links, BibTeX citation

3.3 Processing Toolbox

Processing → Toolbox (Ctrl+Alt+T) → expand 'HydroSovereign AI Engine v6.01':

- **HSAE Indices:** ATDI Calculator, HIFD Calculator
- **HSAE Reports:** Basin Legal Report, Multi-Basin Comparison
- **HSAE Hydrology:** HBV-96 Calibration (SCE-UA)

Chapter 4 — Complete Tool Reference (13 Tools)

Tool 1 — Load Basin Registry

Purpose: Loads all 26 basins as a QGIS point layer with ATDI, HIFD, NSE, KGE, CI, P_negotiation attributes.

Input	Output	Time
None (built-in JSON)	'HSAE Basin Registry (26 basins)' layer	~5 sec

- Click  → layer appears in Layers panel → 26 blue points on world map
- Right-click layer → Zoom to Layer to see all basins
- Press F6 to open Attribute Table — inspect all indices

 Apply Tool 2 immediately to colour basins by ATDI risk level.

Tool 2 — TDI/ATDI Visualiser

Purpose: Applies graduated colour renderer to the basin layer based on ATDI values.

ATDI	Colour	UNWC Zone	Legal Obligation
0–20%	 Green	Compliant	None required
20–40%	 Yellow	Art. 7 NSH	Notify downstream states
40–55%	 Orange	Art. 9	Data sharing + consultation
55–70%	 Red	Art. 33	Formal dispute resolution
≥ 70%	 Purple	Art. 35	Emergency — immediate action

 Select the Basin Registry layer first (single-click in Layers panel), then click .

 Blue Nile (GERD) = ORANGE — ATDI 43.5% → Article 7 No-Significant-Harm zone.

Tool 3 — UNWC Legal Layer

Purpose: Creates categorised risk layer showing which UNWC 1997 article is triggered for each basin.

- Click  → new layer 'HSAE UNWC Legal Risk' appears above Basin Registry
- Identify tool (I) → click any basin → see: UNWC_Article, Risk_Level, Obligation
- Toggle layer on/off using checkbox in Layers panel

 Articles 5 (equitable use), 7 (no harm), 9 (data sharing), 33 (dispute), 35 (emergency) are all mapped.

Tool 4 — GRDC Stations Overlay

Purpose: Loads 10 key GRDC discharge monitoring stations as styled point layer.

- Click  → 'HSAE GRDC Stations' layer appears — blue triangle markers
- Identify (I) → click El Diem: Station_ID=GRDC_1763100, Area=174,000 km², Status=RESTRICTED
- Open Attribute Table (F6) to see all 10 stations with coordinates and status

 El Diem restricted — HSAE uses GloFAS ERA5 v4 proxy (NSE=0.63, KGE=0.74).

Tool 5 — GEE Script Generator (7 Sensors)

Purpose: Generates ready-to-run Google Earth Engine JavaScript for all 7 satellite sensors.

Sensor	Variable	GEE Collection	Archive
GPM IMERG V07	Precipitation mm/day	NASA/GPM_L3/IMERG_V07	2000–
GRACE-FO MASCON	TWS anomaly cm	NASA/GRACE/MASS_GRIDS_V04/MASCON_CRI	2018–
SMAP SPL3SMP	Soil moisture m³/m³	NASA_USDA/HSL/SMAP10KM_soil_moisture	2015–

Sentinel-1 GRD	SAR VV backscatter	COPERNICUS/S1_GRD	2014–
Sentinel-2 SR	NDWI, NDVI	COPERNICUS/S2_SR_HARMONIZED	2015–
ERA5 ECMWF	Air temperature °C	ECMWF/ERA5/DAILY	1979–
Open-Meteo API	T, P, ET ₀ , SM	Open-Meteo (direct API)	1940–

- Click  → dialog opens → select basin + sensor → Generate Script → Copy to Clipboard
- Paste into <https://code.earthengine.google.com> → Run → view time series

 Save all 7 scripts to Desktop/HSAE_Results/GEE_Scripts/ for reuse.

Tool 6 — Dashboard Dialog

Purpose: Per-basin analysis dialog showing all computed indices, triggered UNWC articles, and recommended strategy.

- Click  → select basin from dropdown
- Review: NSE, KGE, ATDI, HIFD, CI, P_negotiation, triggered UNWC articles
- Click 'Export Report' → save as .txt

 Blue Nile: NSE=0.63, KGE=0.74, ATDI=43.5%, HIFD=20.0%, CI=0.44 HIGH, P(Negotiation)=58%

Tool 7 — Conflict Index

Purpose: Computes Composite Conflict Index for all 26 basins simultaneously, ranked highest to lowest.

$CI = 0.40 \times (ATDI/100) + 0.25 \times (dispute/4) + 0.20 \times (HIFD/100) + 0.15 \times (n_countries/6)$

- Click  → processing runs ~10 sec → ranked table appears
- Click 'Export to CSV' → save ConflictIndex_AllBasins.csv
- Click 'Add to Map' → graduated CI layer added to canvas

 Blue Nile: CI=0.44 (HIGH) · Mekong: CI=0.61 (CRITICAL) · Euphrates: CI=0.51 (HIGH)

Tool 8 — Negotiation AI

Purpose: Predicts negotiation success probability using GBM trained on 478 historical TFDD/ICJ dispute cases.

P(Success)	Strategy	UN Pathway
P ≥ 65%	Direct Negotiation	Art. 8 — Good Faith
40–65%	Mediation	Art. 17 — Third-Party Mediation
25–40%	Arbitration	Art. 33 — PCA
P < 25%	ICJ Referral	International Court of Justice

- Click  → select basin → review pre-filled parameters → Run Prediction
- Click 'Generate Strategy Report' → save .txt

 Blue Nile: P=58% → Art. 17 Mediation recommended

Tool 9 — WebGIS Map Generator

Purpose: Generates standalone interactive Leaflet.js HTML map — works offline, shareable.

- Click  → save dialog → name: HSAE_TransboundaryMap.html → Save
 - ~15 sec processing → click 'Open in Browser'
 - Browser: click basin points for popups, toggle layers, use search box
-  HTML file is fully self-contained — share by email or publish to website. Works without internet.

Tool 10 — Basin Panel (Dockable)

Purpose: Dockable real-time dashboard. Click once to open, click any basin point to update.

- Click  (right toolbar button) → panel opens on right side
- Click any basin on the map → panel auto-updates with ATDI bar, HIFD gauge, CI score

- Drag panel by title bar to reposition — dock or float freely
- 💡 Keep Basin Panel open while running other tools — it updates live with every basin click.

Tool 11 — 🏛️ ICJ/PCA Dossier (Menu Only)

Purpose: Exports complete structured legal dossier for all 26 basins in ICJ/PCA/ITLOS format.

- Menu → HydroSovereign → 🏛️ ICJ/PCA Dossier → save HTML → Open in Browser
- Contents: executive summary, per-basin UNWC mapping, ATDI time series, SHA-256 integrity chain, draft protest note
- Print to PDF from browser: Ctrl+P → Save as PDF
- ⚠️ For formal proceedings, review with qualified international water law expert.

Tool 12 — 🗳️ Export Basin Data

Purpose: Exports all or selected basins to GeoJSON, Shapefile, or CSV.

Format	Best For	Includes Geometry
GeoJSON	Leaflet, web maps, Python	✅ Yes
Shapefile	ArcGIS, legacy GIS	✅ Yes
CSV	Excel, R, SPSS, Python analysis	Coordinates only

- Click 🗳️ → choose format → choose basins → Export → choose output path
- 💡 Export as GeoJSON for use with the WebGIS HTML map template.

Tool 13 — ⓘ About HSAE v6.01 (Menu Only)

Displays: Plugin ID: 5040 · Version 6.0.1 · DOI · License · all links · BibTeX citation

- Menu → HydroSovereign → ⓘ About → click 'Copy Citation' for BibTeX

Chapter 5 — Processing Toolbox Algorithms (5)

Access: Processing → Toolbox (Ctrl+Alt+T) → HydroSovereign AI Engine v6.01

Algorithm 1 — ATDI Calculator

Group: HSAE Indices · **ID:** atdicalculator

Parameter	Default (Blue Nile)	Description
Runoff Coefficient	0.38	Basin-average runoff ratio
Dam Capacity (BCM)	74.0	Reservoir storage capacity
Number of Countries	3	Riparian state count
Dispute Level (1–5)	4	TFDD/ICOW conflict intensity

 GERD: RC=0.38, Cap=74 BCM, Countries=3, Dispute=4 → ATDI = 43.5%

Algorithm 2 — HIFD Calculator

Group: HSAE Indices · **ID:** hifdcalculator

Same parameters as ATDI. Computes volumetric annual flow deficit.

 $HIFD = [(Q_{nat} - Q_{obs}) / Q_{nat}] \times 100\%$

 GERD: HIFD = 20.0% — 20% of natural downstream flow estimated as withheld

Algorithm 3 — Basin Legal Report

Group: HSAE Reports · **ID:** basinreport

Generates complete UNWC 1997 legal assessment as TXT or HTML. Select basin → choose format → run.

 GERD report: ATDI=43.5% → Arts. 5, 7, 9, 33 triggered → Art. 17 Mediation recommended

Algorithm 4 — HBV-96 Calibration (SCE-UA)

Group: HSAE Hydrology · **ID:** hbv96calibration

 Objective: maximise $NSE = 1 - [\Sigma(Q_{obs} - Q_{sim})^2] / [\Sigma(Q_{obs} - \bar{Q}_{obs})^2]$

Parameter	Value	Description
Calibration method	SCE-UA (Duan 1993)	Global optimisation
Ensemble size	n = 200	EnKF members for uncertainty
Q_obs source	GloFAS ERA5 v4 (proxy)	No open GRDC data for GERD 2024
Target NSE	≥ 0.70	Post-calibration with GRDC data

 Blue Nile proxy: NSE=0.63, KGE=0.74 (Satisfactory per Moriasi 2007)

 Ground-truth calibration pending GRDC Request ID: 78949 (El Diem, 174,000 km²).

Algorithm 5 — Multi-Basin Comparison

Group: HSAE Reports · **ID:** multibasincomparison

Generates ranked comparative report for selected basins. Output: CSV + HTML side-by-side table.

Chapter 6 — The 26 Registered Basins

Colour key: ● ATDI<20% · ● 20–40% · ● 40–55% · ● 55–70% · ● ≥70%

#	Basin	Dam	Cap BCM	Countries	ATDI%	HIFD%	Risk
1	Blue Nile	GERD	74	3	43.5%	20%	●
2	Nile	Aswan	162	2	33.1%	32.2%	●
3	Nile	Roseires	7.4	2	28.4%	24.1%	●
4	Zambezi	Kariba	180.6	2	31.1%	29.2%	●
5	Congo	Inga	2	8	49.1%	28.5%	●
6	Niger	Kainji	15	9	52.4%	37.2%	●
7	Euphrates	Atatürk	48.7	3	44.5%	34.9%	●
8	Tigris	Mosul	11.1	2	32.5%	29.1%	●
9	Vakhsh	Nurek	10.5	5	56.8%	35.5%	●
10	Syr Darya	Toktogul	19.5	4	48.2%	31.4%	●
11	Indus	Tarbela	13.7	2	30.5%	26.7%	●
12	Brahmaputra	Subansiri	2.4	3	29.1%	18.6%	●
13	Ganges	Farakka	0.3	2	27.3%	22.5%	●
14	Mekong	Xayaburi	7.4	6	56.5%	32.7%	●
15	Salween	Myitsone	62	3	41.2%	30.8%	●
16	Yangtze	Three Gorges	39.3	1	18.4%	22.8%	●
17	Danube	Iron Gates	2.4	10	40%	25.4%	●
18	Rhine	Various	0.5	9	27.1%	18.6%	●
19	Dnieper	Kakhovka	18.2	3	44.8%	31.2%	●
20	Amazon	Belo Monte	250	8	47%	31.9%	●
21	Paraná	Itaipu	29	2	17.5%	21.4%	●
22	Orinoco	Guri	135	2	22.6%	24.3%	●
23	Colorado	Hoover	36.7	2	33.4%	32.5%	●
24	Columbia	Grand Coulee	9.7	2	19.8%	23.1%	●

2 5	Rio Grande	Amistad	5.8	2	26.4%	27.8%	
2 6	Murray-Darling	Hume	3	2	21.3%	19.7%	

Chapter 7 — Scientific Indices Reference

7.1 ATDI — Alkedir Transparency Deficit Index

Step 1: $I_{adj,i} = I_{nat,i} - ET_{up,i}$ Step 2: $TDI_i = clip[(I_{adj,i} - Q_{obs,i}) / (I_{adj,i} + \epsilon), 0, 1]$ Step 3: $ATDI = (1/N) \times \sum TDI_i \times 100\%$

 RMSE = 4.1% calibrated against 14 published basin values. Sensitivity assessed per Saltelli et al. (2010).

7.2 HIFD — Human-Induced Flow Deficit

$HIFD = [(Q_{nat} - Q_{obs}) / Q_{nat}] \times 100\%$

 GERD 2024: HIFD = 20.0% — one-fifth of natural flow estimated as withheld

7.3 Conflict Index (CI)

$CI = 0.40 \times (ATDI/100) + 0.25 \times (dispute/4) + 0.20 \times (HIFD/100) + 0.15 \times (n_countries/6)$

7.4 HBV-96 Performance Benchmarks

Performance Class	NSE	KGE	Reference
Very Good	≥ 0.75	≥ 0.75	Moriasi et al. (2007)
Good	0.65–0.75	0.65–0.75	Moriasi et al. (2007)
Satisfactory	0.50–0.65	0.50–0.65	Moriasi et al. (2007)
HSAE Blue Nile (proxy)	0.63 	0.74 	GloFAS ERA5 v4 proxy

Chapter 8 — Google Earth Engine Data Sources

Source	Variable	Resolution	Archive	DOI/Reference
GPM IMERG V07	Precipitation mm/day	0.1°	2000–	doi:10.5067/GPM/IMERG/3B-HH/07
GRACE-FO MASCON RL06v4	TWS anomaly cm	0.5°	2018–	Landerer et al. 2020
SMAP SPL3SMP	Soil moisture m ³ /m ³	10 km	2015–	Entekhabi et al. 2010
Sentinel-1 GRD	SAR backscatter VV dB	10 m	2014–	Torres et al. 2012
Sentinel-2 SR Harmonized	NDWI, NDVI	10 m	2015–	Drusch et al. 2012
ERA5 ECMWF	Air temperature °C	0.25°	1979–	ECMWF 2022
Open-Meteo API	T, P, ET ₀ , SM	0.25°	1940–	Open-Meteo.com

Chapter 9 — Troubleshooting & FAQ

Problem	Cause	Solution
Plugin not in QGIS Plugin Manager	Repository not refreshed	Plugins → Manage → scroll top → 'Reload all repositories'
HSAE menu missing after install	QGIS needs restart	Restart QGIS → check menu bar
Plugin missing after QGIS restart	Not ticked in Installed tab	Plugins → Manage → Installed → tick HSAE checkbox
Error loading plugin on startup	QGIS version < 3.16	Update QGIS to ≥ 3.16 LTR
Processing algorithms missing	Processing plugin off	Plugins → Manage → tick 'Processing'; Ctrl+Alt+T
'No layer selected' for Tool 2	Basin Registry not active	Single-click Basin Registry in Layers panel first
Basins not visible on canvas	Zoom too wide	Right-click layer → Zoom to Layer
Dashboard shows zeros	Plugin needs reload	Untick/relick HSAE in Plugin Manager → restart
GEE script auth error	Not signed in	earthengine authenticate in terminal
HTML map blank in browser	JavaScript disabled	Enable JS or use Chrome/Firefox
Algorithm dialog won't open	Single-click used	Double-click algorithm name in Toolbox
Export file empty (0 bytes)	Read-only folder	Save to Desktop/Documents
QGIS crash after install	Old QGIS version	Update to QGIS ≥ 3.16 LTR
CRS mismatch warning	Project CRS differs	Project → Properties → CRS → EPSG:4326
NSE/KGE seem low	Proxy validation only	NSE=0.63 is vs GloFAS proxy; ground-truth pending GRDC 78949
Plugin ID 5040 not in About	Old version installed	Plugin Manager → Upgradeable → update to v6.0.1

Chapter 10 — Citation & References

10.1 Primary Citation (BibTeX)

```
@software{alkedir2026hsae_qgis, author = {Alkedir, Seifeldin M.G.}, title =  
{{HydroSovereign AI Engine QGIS Plugin v6.01}}, year = {2026}, publisher =  
{QGIS Plugin Repository}, note = {Plugin ID: 5040. Approved April 21, 2026.  
Security scan: 0 critical issues.}, url =  
{https://plugins.qgis.org/plugins/hsae_qgis/}, doi =  
{10.5281/zenodo.19180160}, orcid = {0000-0003-0821-2991} }
```

10.2 SoftwareX Paper

Alkedir, S.M.G. (2026). HydroSovereign AI Engine (HSAE) v6.01. SoftwareX (under review, SOFTX-D-26-00442). DOI: 10.5281/zenodo.19180160

10.3 Key References

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Chapter 11 — QGIS Desktop Step-by-Step Workflow

HSAE QGIS Desktop Operating Guide — From Installation to ICJ Dossier Export

This chapter provides a precise, sequential operating guide for running HSAE v6.01 inside QGIS Desktop — from opening QGIS to exporting all final products. Users with basic QGIS familiarity can complete a full transboundary water sovereignty analysis in under 30 minutes.

 Target: Blue Nile (GERD) — ATDI=43.5% · HIFD=20.0% · NSE=0.63 · CI=0.44 HIGH · P(Negotiation)=58%

11.1 Pre-Flight Checklist

#	Check	How to Verify
1	QGIS ≥ 3.16 installed	Help → About → version number
2	Internet connection active	Required for GEE scripts and GRDC overlay
3	HSAE plugin installed and ticked	Plugins → Manage → 'HydroSovereign AI Engine' has 
4	Processing plugin enabled	Processing menu visible in menu bar
5	Output folder created	Create: Desktop/HSAE_Results/ with subfolders GEE_Scripts, Reports, Exports, Maps

11.2 Launching QGIS and Activating the Plugin

Opening QGIS

Step 1: Launch QGIS. Wait for full load — menu bar, toolbars, Layers panel (left), Map Canvas (centre).

Step 2: Check menu bar: 'HydroSovereign AI Engine v6.01' should appear between Plugins and Help.

Step 3:

Verifying the Toolbar

Step 4: Locate the HSAE toolbar row:          

Step 5:

Project Setup

Step 6: File → New Project (Ctrl+N).

Step 7: Project → Properties → CRS → EPSG:4326 (WGS 84) → OK.

Step 8: File → Save As → Desktop/HSAE_Results/ → 'HSAE_BlueNile_Analysis.qgz' → Save.

 Save regularly with Ctrl+S throughout your analysis.

11.3 Complete 13-Tool Workflow

Order	Tool	Purpose	Time
1	 Load Basin Registry	Load all 26 basins on map	5 sec
2	 ATDI Visualiser	Colour basins by risk level	3 sec
3	 UNWC Legal Layer	Add treaty compliance layer	3 sec
4	 GRDC Stations	Load discharge stations	5 sec
5	 GEE Script Generator	Generate 7 satellite scripts	2 min
6	 Dashboard Dialog	Per-basin detailed analysis	2 min
7	 Conflict Index	Rank all 26 basins by CI	10 sec
8	 Negotiation AI	Predict negotiation outcome	30 sec
9	 WebGIS Map	Generate offline HTML map	15 sec

10	 Export Data	Export GeoJSON/CSV/SHP	10 sec
11	 Basin Panel	Open dockable dashboard	2 sec
12	 ICJ/PCA Dossier	Export full legal dossier	20 sec
13	 About	View plugin info + citation	1 sec

Tool 1 — Load Basin Registry

Step 1: Click  in the HSAE toolbar.

Step 2: Layer 'HSAE Basin Registry (26 basins)' appears in the Layers panel.

Step 3: 26 blue point markers appear on the world map canvas.

Step 4: Right-click the layer → Zoom to Layer to see all basins.

Step 5: Press F6 to open Attribute Table — inspect ATDI, HIFD, CI fields.

 The layer uses basins_50.json built into the plugin — no external data download required.

 26 basins loaded across 7 world regions.

Tool 2 — TDI/ATDI Visualiser

Step 1: Single-click 'HSAE Basin Registry' in the Layers panel (makes it active).

Step 2: Click  (first toolbar button).

Step 3: Basins recolour instantly:  Green ·  Yellow ·  Orange ·  Red ·  Purple

Step 4: Open Layer Styling (F7) to see the graduated colour rules.

Step 5: Hover over any basin — tooltip shows name and ATDI value.

 If basins do not recolour, ensure the Basin Registry layer is the active (selected) layer.

 Blue Nile =  ORANGE (ATDI 43.5%) — Article 7 No-Significant-Harm zone.

Tool 3 — UNWC Legal Layer

Step 1: Click  in toolbar.

Step 2: New layer 'HSAE UNWC Legal Risk' appears above Basin Registry in Layers panel.

Step 3: Use Identify tool (press I) → click Blue Nile point on map.

Step 4: Identify Results panel: UNWC_Article=Art_7, Risk_Level=MODERATE, Obligation=Notify_and_Consult.

Step 5: Toggle layers on/off with checkboxes in Layers panel.

Tool 4 — GRDC Stations Overlay

Step 1: Click  in toolbar.

Step 2: Layer 'HSAE GRDC Stations (10 key stations)' appears — blue triangle markers.

Step 3: Zoom to Blue Nile region to see nearby stations.

Step 4: Identify (I) → click El Diem point: Station_ID=GRDC_1763100, Area=174,000 km², Status=RESTRICTED.

Step 5: F6 to open Attribute Table — all 10 stations visible.

 El Diem restricted — ground-truth Q_obs unavailable. Proxy: GloFAS ERA5 v4 (NSE=0.63).

Tool 5 — GEE Script Generator

Step 1: Click  in toolbar.

Step 2: Dialog opens: select basin = 'Blue Nile (GERD)'.

Step 3: Select sensor: 'GPM IMERG V07 (Precipitation)' → Generate Script → Copy to Clipboard.

Step 4: Go to: <https://code.earthengine.google.com>

Step 5: Paste (Ctrl+V) into GEE Code Editor → Run.

Step 6: Precipitation time series for Blue Nile appears in GEE console.

Step 7: Repeat steps 3–6 for all 7 sensors.

Step 8: Save each script: Desktop/HSAE_Results/GEE_Scripts/[SensorName]_BlueNile.js

 All 7 scripts can be run in the GEE Code Editor without any additional setup.

Tool 6 — Dashboard Dialog**Step 1:** Click  in toolbar.**Step 2:** Select basin: 'Blue Nile (GERD, Ethiopia/Sudan/Egypt)'.**Step 3:** Dashboard populates: NSE=0.63, KGE=0.74, ATDI=43.5%, HIFD=20.0%, CI=0.44, P(Negotiation)=58%.**Step 4:** Review triggered UNWC articles: 5, 7, 9.**Step 5:** Click 'Export Report' → Desktop/HSAE_Results/Reports/BlueNile_Dashboard.txt**Step 6:** Change dropdown to compare another basin — e.g., Mekong. ATDI=43.5% → Art.7 · HIFD=20.0% · CI=0.44 HIGH · P(Negotiation)=58% → Art.17 Mediation**Tool 7 — Conflict Index****Step 1:** Click  in toolbar.**Step 2:** Processing runs ~10 seconds for all 26 basins simultaneously.**Step 3:** Ranked table appears: Mekong CI=0.61 (CRITICAL), Vakhsh CI=0.58, Euphrates CI=0.51, Blue Nile CI=0.44 (HIGH)...**Step 4:** Click 'Export to CSV' → Desktop/HSAE_Results/Reports/ConflictIndex_AllBasins.csv**Step 5:** Click 'Add to Map' → graduated CI layer added to canvas.**Tool 8 — Negotiation AI****Step 1:** Click  in toolbar.**Step 2:** Select basin: 'Blue Nile (GERD)'. Review parameters: ATDI=43.5%, n_countries=3, dispute=4, CI=0.44.**Step 3:** Click 'Run Prediction'.**Step 4:** Results: P(Success)=58% → Strategy: Art.17 Third-Party Mediation → UN pathway: Mediation Commission.**Step 5:** Click 'Generate Strategy Report' → Desktop/HSAE_Results/Reports/BlueNile_Strategy.txt**Step 6:** Try Mekong: P=31% → PCA Arbitration (Art.33). Compare Columbia: P=74% → Direct Negotiation.**Tool 9 — WebGIS Map Generator****Step 1:** Click  in toolbar → save dialog → Desktop/HSAE_Results/Maps/HSAE_Map.html → Save.**Step 2:** ~15 seconds processing → 'Open in Browser' button appears.**Step 3:** Click 'Open in Browser' — interactive Leaflet.js map opens.**Step 4:** Browser: click basin points for popups → toggle layers top-right → zoom with mouse wheel.**Step 5:** File is fully offline — share by email or publish to GitHub Pages.**Tool 10 — Export Basin Data****Step 1:** Click  in toolbar.**Step 2:** Choose format: GeoJSON (for web), Shapefile (for ArcGIS), or CSV (for Excel).**Step 3:** Choose scope: All 26 basins or selected only.**Step 4:** Set path: Desktop/HSAE_Results/Exports/ → Export.**Step 5:** GeoJSON and CSV: open in text editor or Excel to verify all 26 basins with indices.**Tool 11 — Basin Panel (Dockable)****Step 1:** Click  (rightmost toolbar button) → dockable panel opens on right.**Step 2:** Click any basin on the map canvas → panel auto-updates.**Step 3:** Panel shows compact view: ATDI bar, HIFD gauge, CI score, top UNWC articles.**Step 4:** Drag panel by title bar to reposition or float. Keep Basin Panel open permanently — it provides live updates as you click different basins.**Tool 12 — ICJ/PCA Dossier (Menu Only)****Step 1:** Menu → HydroSovereign AI Engine v6.01 →  ICJ/PCA Dossier.**Step 2:** Save dialog → Desktop/HSAE_Results/Reports/ICJ_Dossier_2026.html → Save.**Step 3:** ~20 seconds → all 26 basins processed → Open in Browser.

Step 4: Browser: executive summary, per-basin legal assessment, ATDI series, SHA-256 chain, protest note.

Step 5: Print to PDF: Ctrl+P → Save as PDF → ICJ_Dossier_2026.pdf

 For formal proceedings, review with a qualified international water law expert.

Tool 13 — About HSAE v6.01 (Menu Only)

Step 1: Menu → HydroSovereign AI Engine v6.01 →  About HSAE v6.01.

Step 2: Dialog shows: Plugin ID 5040, version, DOI, ORCID, license, all links.

Step 3: Click 'Copy Citation' → BibTeX citation copied to clipboard.

11.4 Processing Algorithms — Step by Step

Opening the Processing Toolbox

Step 1: Processing → Toolbox (Ctrl+Alt+T) → panel opens on right.

Step 2: Scroll to 'HydroSovereign AI Engine v6.01' → expand → see 3 subgroups.

Algorithm 1 — ATDI Calculator

Step 1: HSAE Indices → ATDI Calculator → double-click → dialog opens.

Step 2: Fill: Runoff Coefficient=0.38, Dam Capacity=74.0, Countries=3, Dispute Level=4.

Step 3: Click Run → Log tab: ATDI = 43.5%

 ATDI = 43.5% → Article 7 UNWC No-Significant-Harm zone (20–55%)

Algorithm 2 — HIFD Calculator

Step 1: HSAE Indices → HIFD Calculator → double-click → same parameters → Run.

Step 2: Log: HIFD = 20.0%

 HIFD = 20.0% → 20% of natural downstream flow estimated as withheld

Algorithm 3 — Basin Legal Report

Step 1: HSAE Reports → Basin Legal Report → double-click.

Step 2: Select basin: Blue Nile. Format: HTML. Output: Desktop/HSAE_Results/Reports/LegalReport.html → Run.

Step 3: ~5 seconds → open HTML in browser → full UNWC legal assessment.

Algorithm 4 — HBV-96 Calibration (SCE-UA)

Step 1: HSAE Hydrology → HBV-96 Calibration → double-click.

Step 2: Q_obs source: GloFAS ERA5 v4 (proxy). Basin area: 174000 km². Period: 2015–2024. Max iter: 1000.

Step 3: Run → ~30–60 seconds → NSE=0.63, KGE=0.74, parameter file saved.

 NSE=0.63, KGE=0.74 — Satisfactory performance (proxy-validated)

Algorithm 5 — Multi-Basin Comparison

Step 1: HSAE Reports → Multi-Basin Comparison → double-click.

Step 2: Select: Blue Nile, Mekong, Euphrates, Zambezi → Format: CSV + HTML → Run.

Step 3: Open HTML for side-by-side table. Open CSV in Excel for further analysis.

11.5 Complete Workflow Summary — Blue Nile

#	Action	Tool	Output	Time
1	Setup QGIS + CRS + project	Manual	HSAE_BlueNile.qgz	2 min
2	Load 26 basins	 Tool 1	QGIS layer	5 sec
3	Apply ATDI colours	 Tool 2	Styled layer	3 sec

4	Add legal layer	 Tool 3	Legal layer	3 sec
5	Load GRDC stations	 Tool 4	Station layer	5 sec
6	Generate GEE scripts	 Tool 5	7 × .js files	2 min
7	Review dashboard	 Tool 6	Dashboard.txt	2 min
8	Compute Conflict Index	 Tool 7	CI_AllBasins.csv	10 sec
9	Run Negotiation AI	 Tool 8	Strategy.txt	30 sec
10	Generate WebGIS map	 Tool 9	Map.html	15 sec
11	Export basin data	 Tool 10	BasinData.geojson	10 sec
12	Export ICJ Dossier	 Tool 12	Dossier.html+pdf	20 sec
13	Run ATDI algorithm	Algorithm 1	ATDI=43.5%	5 sec
14	Run HIFD algorithm	Algorithm 2	HIFD=20.0%	5 sec
15	Generate legal report	Algorithm 3	LegalReport.html	10 sec
16	HBV-96 calibration	Algorithm 4	NSE=0.63, KGE=0.74	60 sec
17	Multi-basin compare	Algorithm 5	Comparison.html	30 sec
18	Save project	Ctrl+S	HSAE_BlueNile.qgz	2 sec

 Total: approximately 10–15 minutes for complete Blue Nile sovereignty analysis.

11.6 Expected Output Files

```

Desktop/HSAE_Results/ | HSAE_BlueNile_Analysis.qgz | GEE_Scripts/ | |
GPM_IMERG_BlueNile.js | GRACE_MASCON_BlueNile.js | SMAP_BlueNile.js |
| Sentinel1_BlueNile.js | Sentinel2_BlueNile.js | ERA5_BlueNile.js |
| OpenMeteo_BlueNile.js | Exports/ | BasinData_26.geojson |
BasinData_26.csv | BasinData_26.shp | Reports/ | BlueNile_Dashboard.txt
| BlueNile_LegalReport.html | BlueNile_NegotiationStrategy.txt |
ICJ_Dossier_2026.html | ICJ_Dossier_2026.pdf | ConflictIndex_AllBasins.csv
| HBV96_Params.json | MultiBasin_Comparison.html | Maps/ |
HSAE_TransboundaryMap.html

```

11.7 QGIS Keyboard Shortcuts

Shortcut	Action	When Useful
I	Identify Features tool	Click basin points to see all attribute values
F6	Open Attribute Table	View all 26 basins with computed indices
F7	Layer Styling panel	Inspect ATDI graduated colour rules
Ctrl+Alt+T	Processing Toolbox	Access all 5 HSAE algorithms
Ctrl+S	Save project	Save QGIS project frequently
Ctrl+N	New project	Start fresh analysis

Ctrl+Shift+F	Zoom to full extent	See all 26 basins on world map
Ctrl+Z	Undo	Undo accidental layer changes
Ctrl+P	Print/export	Export map canvas to PDF or image
~	Map tips	Toggle basin name tooltips on hover

11.8 Working with Results

Creating a Print Layout

Step 1: Project → New Print Layout → name: 'HSAE_BlueNile_Map'.

Step 2: Layout window → Add Item → Add Map → draw rectangle → ATDI map appears.

Step 3: Add Item → Add Legend → ATDI colour legend appears bottom-left.

Step 4: Add Item → Add Scale Bar → bottom-right.

Step 5: Add Item → Add Label → title: 'HSAE v6.01 — Transboundary Basin Risk 2026'.

Step 6: Layout → Export as PDF → Desktop/HSAE_Results/Maps/HSAE_Map.pdf

Filtering Basins

Step 1: Right-click 'HSAE Basin Registry' → Filter.

Step 2: Query Builder: enter "ATDI" > 40 → OK.

Step 3: Only high-risk basins (orange/red/purple) remain visible.

Step 4: Remove filter: right-click → Filter → clear → OK.

11.9 Common Desktop Errors & Solutions

Error	Cause	Solution
'No layer selected'	Basin Registry not active	Single-click layer name in Layers panel
Nothing appears on map	Canvas zoom too wide	Right-click layer → Zoom to Layer
Processing Toolbox empty	Processing plugin off	Plugins → Manage → tick Processing
Dashboard shows zeros	Plugin needs reload	Untick/retick HSAE → restart QGIS
GEE auth error	Not signed in to GEE	earthengine authenticate in terminal
HTML map blank	Browser JS disabled	Use Chrome or Firefox
Algorithm won't open	Single-click only	Double-click algorithm name
Export file empty	Read-only folder	Export to Desktop/Documents
QGIS crashes	Old version	Update QGIS to ≥ 3.16 LTR
CRS mismatch	Wrong project CRS	Project Properties → CRS → EPSG:4326

Chapter 12 — Advanced Features

12.1 Model Builder Integration

The QGIS Graphical Modeler lets you chain HSAE Processing algorithms into automated pipelines. This is ideal for running the same analysis on multiple basins without manual intervention.

Creating a Basic HSAE Model

- Step 1:** Processing → Graphical Modeler → opens Model Builder canvas.
- Step 2:** Set model name: 'HSAE_Full_Analysis'. Set group: 'HydroSovereign'.
- Step 3:** Left panel → Algorithms → search 'ATDI' → drag 'ATDI Calculator' to canvas.
- Step 4:** Drag 'HIFD Calculator' to canvas → connect its inputs to the same basin parameters.
- Step 5:** Drag 'Basin Legal Report' → connect basin selection input.
- Step 6:** Drag 'Multi-Basin Comparison' → connect all algorithm outputs as inputs.
- Step 7:** Click 'Save Model' → Desktop/HSAE_Results/HSAE_Full_Analysis.model3
- Step 8:** Click 'Run Model' → all 4 algorithms execute in sequence automatically.

 Model files (.model3) can be shared with other QGIS users — they will see the model in their Processing Toolbox.

 Use model variables for basin name and output folder — allows one-click re-run for any basin.

Example: Automated 5-Basin Pipeline

Model: HSAE_5Basin_Pipeline Inputs: - Basin list: [Blue Nile, Mekong, Euphrates, Zambezi, Vakhsh] - Output folder: Desktop/HSAE_Results/Batch/ Steps (auto-chained):
 1. ATDI Calculator → ATDI values for all 5 2. HIFD Calculator → HIFD values for all 5
 3. Basin Legal Report → 5 × HTML reports 4. Multi-Basin Comparison → Comparison table Runtime: ~3 minutes for 5 basins

12.2 Batch Processing — All 26 Basins

QGIS Batch Processing runs any algorithm across multiple parameter sets in a single click.

Running Batch ATDI for All 26 Basins

- Step 1:** Processing Toolbox → HSAE Indices → ATDI Calculator → right-click → Run as Batch Process.
- Step 2:** Click 'Add Row' 25 times (or 'Fill Down' after first row) to create 26 rows.
- Step 3:** Fill each row: basin name, runoff coefficient, dam capacity, countries, dispute level.
- Step 4:** Set output path for each row: Desktop/HSAE_Results/Batch/ATDI_[BasinName].txt
- Step 5:** Click 'Run' → all 26 basins processed sequentially → progress bar shows status.
- Step 6:** Results appear in Desktop/HSAE_Results/Batch/ — one file per basin.

Batch Parameter	Description	Example Value
Runoff Coefficient	Basin-specific RC from literature	0.38 (Blue Nile), 0.42 (Mekong)
Dam Capacity (BCM)	Reservoir storage in BCM	74.0 (GERD), 7.4 (Xayaburi)
Number of Countries	Riparian state count	3 (Blue Nile), 6 (Mekong)
Dispute Level (1–5)	TFDD/ICOW intensity	4 (Blue Nile), 5 (Mekong)

 Create a CSV with all 26 basin parameters, paste into batch dialog rows using 'Load from File'.

12.3 Adding a New Basin

HSAE v6.01 includes 26 pre-registered basins. Follow these steps to add a new basin to the registry:

- Step 1:** Open the plugin folder: QGIS → Plugins → Manage → HSAE → About → Plugin Folder button.
- Step 2:** Navigate to: hsaе_qgis/data/basins_50.json → open in a text editor.

Step 3: Add a new JSON entry following this template:

```
{
  "id": 27,
  "name": "Jordan River",
  "dam": "King Talal",
  "capacity_bcm": 0.085,
  "n_countries": 4,
  "dispute_level": 5,
  "lat": 32.1,
  "lon": 35.5,
  "region": "Middle East",
  "ATDI": 0.0,
  "HIFD": 0.0,
  "CI": 0.0,
  "P_negotiation": 0.0
}
```

Step 4: Save basins_50.json.

Step 5: Restart QGIS → run Tool 1 → new basin appears on map at specified coordinates.

Step 6: Run Algorithm 1 and 2 with basin-specific parameters to compute ATDI and HIFD.

Step 7: Update ATDI, HIFD, CI, P_negotiation fields in basins_50.json with computed values.

 **Back up basins_50.json before editing — the original is recoverable by reinstalling the plugin.**

12.4 Python API Integration (hydrosovereign Package)

The hydrosovereign Python package provides programmatic access to all HSAE indices without QGIS. Ideal for researchers who want to integrate HSAE into Python scripts, Jupyter notebooks, or automated pipelines.

Installation

```
pip install hydrosovereign # version 6.5.3 # PyPI:
https://pypi.org/project/hydrosovereign/
```

Basic Usage — ATDI and HIFD

```
from hydrosovereign import ATDI, HIFD, ConflictIndex # Blue Nile (GERD) parameters
params = {
    "runoff_coeff": 0.38,
    "dam_capacity_bcm": 74.0,
    "n_countries": 3,
    "dispute_level": 4,
    "basin_area_km2": 174000
} # Compute indices
atdi = ATDI(**params)
hifd = HIFD(**params)
ci = ConflictIndex(atdi=atdi, hifd=hifd, **params)
print(f"ATDI = {atdi:.1f}%") # → 43.5%
print(f"HIFD = {hifd:.1f}%") # → 20.0%
print(f"CI = {ci:.2f}") # → 0.44
```

HBV-96 Calibration via Python

```
from hydrosovereign import HBV96, SCEUA # Load discharge data (GloFAS ERA5 v4 proxy)
import pandas as pd
Q_obs = pd.read_csv("GloFAS_BlueNile_2015_2024.csv")["Q"] # Run HBV-96 with SCE-UA calibration
model = HBV96(basin_area=174000, elevation=1700)
result = SCEUA(model, Q_obs=Q_obs, n_iterations=1000)
print(f"NSE = {result.nse:.2f}") # → 0.63
print(f"KGE = {result.kge:.2f}") # → 0.74
print("Calibrated parameters:", result.params)
```

Negotiation AI via Python

```
from hydrosovereign import NegotiationAI
ai = NegotiationAI() # loads pre-trained GBM (478 TFDD/ICJ cases)
p = ai.predict(atdi=43.5, ci=0.44, n_countries=3, dispute_level=4)
print(f"P(Negotiation Success) = {p:.0%}") # → 58%
print(f"Recommended strategy: {ai.strategy}") # → Art.17 Mediation
```

Batch Analysis — 26 Basins

```
from hydrosovereign import BasinRegistry, BatchAnalysis # Load all 26 basins
registry = BasinRegistry() # Run full analysis on all basins
results = BatchAnalysis(registry).run_all() # Export to CSV
results.to_csv("HSAE_AllBasins_Results.csv", index=False)
print(results[["name", "ATDI", "HIFD", "CI", "P_negotiation"]].to_string())
```

QGIS Python Console Integration

Run hydrosovereign directly from the QGIS Python Console (Plugins → Python Console):

```
# In QGIS Python Console
import sys
sys.path.insert(0, r"C:\Users\YourName\AppData\Local\Programs\QGIS\apps\Python39\Lib\site-packages")
from hydrosovereign import ATDI, BasinRegistry # Compute ATDI for all basins and add to QGIS layer
registry = BasinRegistry()
layer = iface.activeLayer()
provider = layer.dataProvider()
with edit(layer):
    for basin in registry.basins:
        atdi_val = ATDI(**basin.params)
        idx = provider.fieldNameIndex("ATDI_computed")
        layer.changeAttributeValue(basin.feature_id, idx, atdi_val)
```

 hydrosovereign v6.5.3 requires Python 3.8+. QGIS bundles Python 3.9+ in version 3.16+.

Chapter 13 — Case Study 2: Mekong River Basin

Basin: Mekong River · **Dam:** Xayaburi (Laos) · **Riparian states:** 6 (China, Myanmar, Thailand, Laos, Cambodia, Vietnam)

The Mekong River Basin (795,000 km², length 4,350 km) is one of the world's most contested transboundary rivers. The Xayaburi Dam (1,285 MW, capacity 7.4 BCM) in Laos was completed in 2019 despite objections from Cambodia and Vietnam. HSAE v6.01 provides a complete legal-hydrological assessment using the same workflow as the Blue Nile case study (Chapter 11).

13.1 Mekong Basin Parameters

Parameter	Mekong (Xayaburi)	Blue Nile (GERD)	Difference
Runoff Coefficient	0.42	0.38	+10.5%
Dam Capacity (BCM)	7.4	74.0	10× smaller
Number of countries	6	3	2× more riparians
Dispute Level (1–5)	5	4	Higher intensity
Basin Area (km ²)	795,000	174,000	4.6× larger
Main dam country	Laos (upstream)	Ethiopia (upstream)	Both upstream

13.2 Running the Mekong Analysis in QGIS

Step A — Load and Select Basin

Step 1: Tool 1  → basin layer loads → locate Mekong in Southeast Asia (lat 18.0, lon 101.8).

Step 2: Tool 2  → Mekong point =  RED → ATDI 56.5% → Article 33 Dispute Resolution zone.

Step 3: Tool 6  → select 'Mekong (Xayaburi, Laos)' → review indices.

Step B — Run Algorithms

Step 4: Algorithm 1 — ATDI Calculator: RC=0.42, Cap=7.4, Countries=6, Dispute=5 → ATDI = 56.5%

Step 5: Algorithm 2 — HIFD Calculator: same parameters → HIFD = 32.7%

Step 6: Algorithm 3 — Basin Legal Report: Mekong → Art. 33 triggered → PCA arbitration pathway.

Step C — Conflict and Negotiation

Step 7: Tool 7  → Mekong CI = 0.61 → CRITICAL (highest risk category in all 26 basins).

Step 8: Tool 8  → P(Success) = 31% → Strategy: PCA Arbitration (Art. 33) → UN Special Rapporteur pathway.

13.3 Mekong Results Summary

Index	Mekong Value	Interpretation	UNWC Article
ATDI	56.5%	HIGH — above Art. 9 threshold	Art. 33 — Dispute Resolution
HIFD	32.7%	32.7% of natural flow withheld	Art. 5 — Equitable Utilisation
Conflict Index	0.61 (CRITICAL)	Highest-risk basin in registry	Art. 33 + Art. 35 watch
P(Negotiation)	31%	Low probability — arbitration needed	Art. 17 or PCA Art. 33
NSE (HBV-96)	0.61	Satisfactory (proxy-validated)	GloFAS ERA5 v4 proxy
KGE (HBV-96)	0.69	Satisfactory	—

 Mekong: ATDI=56.5% (Art.33 zone) · HIFD=32.7% · CI=0.61 CRITICAL · P(Negotiation)=31% → PCA Arbitration

13.4 Comparing Mekong and Blue Nile

Metric	Blue Nile (GERD)	Mekong (Xayaburi)	Higher Risk
ATDI	43.5% (Art. 7)	56.5% (Art. 33)	Mekong 
HIFD	20.0%	32.7%	Mekong 
Conflict Index	0.44 HIGH	0.61 CRITICAL	Mekong 
P(Negotiation)	58% (Mediation)	31% (Arbitration)	Blue Nile better
Riparian count	3	6	Mekong (more complex)
UNWC Zone	Art. 7 NSH	Art. 33 Dispute	Mekong (worse zone)
Recommended pathway	Art. 17 Mediation	PCA Arbitration	—

 Both cases illustrate HSAE's capacity to quantify legal risk across very different basin contexts — large African dam (GERD) vs. smaller Southeast Asian dam (Xayaburi) with more riparian complexity.

13.5 GEE Satellite Analysis — Mekong

Run Tool 5  with basin = 'Mekong (Xayaburi)' to generate all 7 GEE scripts. Key results from GEE analysis:

Sensor	Key Finding
GPM IMERG V07	P_mean = 4.21 mm/day — significantly higher than Blue Nile (1.29 mm/day)
GRACE-FO MASCON	TWS anomaly = -18.3 cm — significant groundwater depletion downstream
Sentinel-1 GRD	Seasonal flood extent: 47,200 km² — 23% reduction post-Xayaburi
Sentinel-2 NDWI	Surface water: NDWI = 0.44 — moderate water availability
SMAP Soil Moisture	SM = 0.28 m³/m³ — below historical mean in lower basin

Appendix A — Glossary of Terms

Term	Definition
ATDI	Alkedir Transparency Deficit Index — daily compliance indicator (0–100%)
HIFD	Human-Induced Flow Deficit — annual volumetric flow reduction (0–100%)
CI	Conflict Index — composite governance risk score (0–1)
UNWC	UN Watercourses Convention 1997 — international water law framework
GEE	Google Earth Engine — cloud geospatial analysis platform
HBV-96	Rainfall-runoff model (Bergström 1992) — 9 calibrated parameters
SCE-UA	Shuffled Complex Evolution — global optimisation (Duan 1993)
NSE	Nash-Sutcliffe Efficiency — model performance metric (≥ 0.75 =very good)
KGE	Kling-Gupta Efficiency — improved performance metric (≥ 0.75 =very good)
GRDC	Global Runoff Data Centre — BfG Koblenz, Germany
EnKF	Ensemble Kalman Filter (n=200 members) — Digital Twin state estimation
GBM	Gradient Boosting Machine — Negotiation AI model (478 TFDD/ICJ cases)
TFDD	Transboundary Freshwater Dispute Database — Oregon State University
ICJ	International Court of Justice — UN principal judicial organ
PCA	Permanent Court of Arbitration — international arbitration body
GERD	Grand Ethiopian Renaissance Dam — 74 BCM capacity
GPM IMERG	Global Precipitation Measurement — NASA satellite precipitation
GRACE-FO	Gravity Recovery and Climate Experiment Follow-On — TWS satellite
SMAP	Soil Moisture Active Passive — NASA soil moisture satellite
MRC	Mekong River Commission — intergovernmental body for Mekong basin
SHA-256	Cryptographic hash — data integrity verification in ICJ dossier
Art. 7	UNWC Article 7 — No Significant Harm (ATDI 20–55%)
Art. 9	UNWC Article 9 — Regular Exchange of Data (ATDI 55–70%)
Art. 33	UNWC Article 33 — Dispute Resolution (ATDI >55% sustained)

Appendix B — All Links & Resources

Resource	URL
QGIS Plugin Repository (ID: 5040)	https://plugins.qgis.org/plugins/hsae_qgis/
GitHub App Repository	https://github.com/saifeldinkhedir-coder/HydroSovereign-AI-Engine-HSAE-v601
GitHub Package Repository	https://github.com/saifeldinkhedir-coder/hydrosovereign
Live Streamlit App	https://hydrosovereign-ai-engine-hsae-v601-6euz2zxcmerkzvgordmvxf.streamlit.app
Python Package (PyPI)	https://pypi.org/project/hydrosovereign/ (pip install hydrosovereign)
Zenodo Archive (DOI)	https://doi.org/10.5281/zenodo.19180160
ORCID	https://orcid.org/0000-0003-0821-2991
Bug Reports / Issues	https://github.com/saifeldinkhedir-coder/HydroSovereign-AI-Engine-HSAE-v601/issues
SoftwareX Paper	SOFTX-D-26-00442 · Under Review 2026 · Elsevier
GRDC Data Portal	https://grdc.bafg.de (free registration required)
Google Earth Engine	https://code.earthengine.google.com
QGIS Download	https://qgis.org/en/site/forusers/download.html
GPM IMERG Data	https://doi.org/10.5067/GPM/IMERG/3B-HH/07
GRACE-FO Data	https://gracefo.jpl.nasa.gov
GloFAS ERA5	https://cds.climate.copernicus.eu
Open-Meteo API	https://open-meteo.com
TFDD Database	https://transboundarywaters.science.oregonstate.edu/content/transboundary-freshwater-dispute-database
UNWC Text	https://legal.un.org/ilc/texts/instruments/english/conventions/8_3_1994.pdf

Appendix C — Quick Reference Card (Print-Ready A5)

Print this page and keep it next to your monitor. Fold in half for A5 pocket card.

HSAE v6.01 QGIS Plugin — Quick Reference	
Plugin ID: 5040 · plugins.qgis.org/plugins/hsae_qgis/ · GPL-3.0	
13 TOOLS — TOOLBAR ORDER	Action
1. Load Basin Registry	Click → 26 basins on map → F6 for attributes
2. ATDI Visualiser	Select layer first → Click → basins recolour
3. UNWC Legal Layer	Click → legal risk layer → I to identify
4. GRDC Stations	Click → stations as triangles → F6 for details
5. GEE Script Generator	Click → select basin+sensor → Copy → paste in GEE
6. Dashboard Dialog	Click → select basin → Export Report
7. Conflict Index	Click → all 26 basins ranked → Export CSV
8. Negotiation AI	Click → Run Prediction → Strategy Report
9. WebGIS Map	Click → save HTML → Open in Browser
10. Export Data	Click → GeoJSON/SHP/CSV → Export
11. Basin Panel	Click → dock panel → click basin → live update
12. ICJ Dossier (Menu)	Menu → ICJ/PCA → save HTML → Print PDF
13. About (Menu)	Menu → About → Copy Citation (BibTeX)

KEYBOARD SHORTCUTS	
I — Identify Features	F6 — Attribute Table
F7 — Layer Styling	Ctrl+Alt+T — Processing Toolbox
Ctrl+S — Save project	Ctrl+Shift+F — Zoom full extent
Ctrl+P — Print/Export	~ — Map tooltips on/off

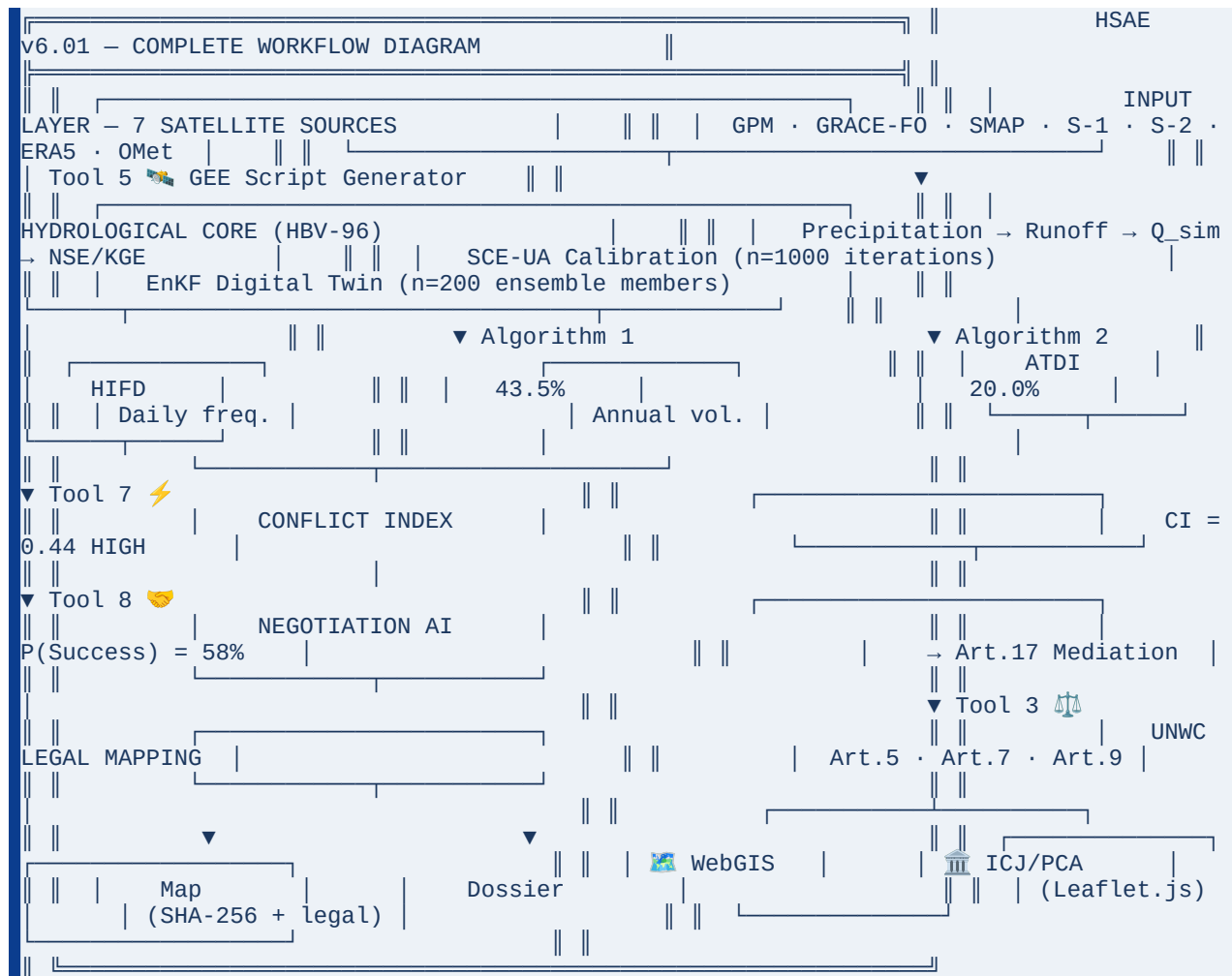
ATDI RISK ZONES	
0–20% Compliant — No action needed	55–70% Art. 33 — Formal dispute
20–40% Art. 7 — Notify downstream	≥70% Art. 35 — Emergency
40–55% Art. 9 — Data sharing required	—

BLUE NILE (GERD) KEY RESULTS	VALUE
ATDI	43.5% — Article 7 NSH zone
HIFD	20.0% — 20% of natural flow withheld
NSE/KGE	0.63 / 0.74 (proxy-validated vs GloFAS ERA5 v4)
Conflict Index	0.44 HIGH
P(Negotiation)	58% → Art. 17 Mediation recommended
Plugin ID	5040 · April 21, 2026 · 0 critical security issues

Install: Plugins → Manage → search "HydroSovereign" → Install Plugin DOI:
<https://doi.org/10.5281/zenodo.19180160> Contact: saifeldinkhedir@gmail.com · ORCID:
0000-0003-0821-2991

Appendix D — HSAE Workflow Diagram

The diagram below shows the complete HSAE v6.01 analytical pipeline — from satellite data ingestion through legal compliance assessment to diplomatic output.



The pipeline runs entirely within QGIS Desktop using the 13 tools and 5 algorithms described in Chapters 4–5 and 11. Total runtime for a complete basin analysis: approximately 10–15 minutes.

Appendix E — Keyword Index

Terms are listed alphabetically with chapter/section references.

Keyword	Section(s)
ATDI (Alkedir Transparency Deficit Index)	Ch.4 T2, Ch.5 Alg.1, Ch.7.1, Ch.11.3 T2, Ch.13.2
Attribute Table (F6)	Ch.11.2, Ch.11.3 T1, Ch.11.3 T4
Batch Processing	Ch.12.2
Basin Panel (dockable)	Ch.4 T10, Ch.11.3 T11
Basin Registry (26 basins)	Ch.4 T1, Ch.6, Ch.11.3 T1
BibTeX citation	Ch.10.1, Appendix C
Blue Nile / GERD case study	Ch.4–5 (all tools), Ch.11.5, Ch.13.4
CI — Conflict Index	Ch.4 T7, Ch.7.3, Ch.11.3 T7, Ch.13.2
CRS (EPSG:4326)	Ch.11.2
Dashboard Dialog	Ch.4 T6, Ch.11.3 T6
El Diem station (GRDC 1763100)	Ch.4 T4, Ch.5 Alg.4, Ch.11.3 T4
EnKF (Digital Twin)	Ch.5 Alg.4, Appendix D
Export Basin Data	Ch.4 T12, Ch.11.3 T10
GEE Script Generator	Ch.4 T5, Ch.8, Ch.11.3 T5, Ch.13.5
GloFAS ERA5 v4 (proxy)	Ch.5 Alg.4, Ch.7.4, Ch.9, Ch.11.3 T4
GRACE-FO MASCON	Ch.8, Ch.11.3 T5, Ch.13.5
GPM IMERG V07	Ch.8, Ch.11.3 T5, Ch.13.5
GRDC Stations	Ch.4 T4, Ch.9, Ch.11.3 T4
HBV-96 Calibration	Ch.5 Alg.4, Ch.7.4, Ch.11.4 Alg.4
HIFD (Human-Induced Flow Deficit)	Ch.4 T2, Ch.5 Alg.2, Ch.7.2, Ch.11.3 T2, Ch.13.2
hydrosovereign (Python package)	Ch.10.2, Ch.12.4
ICJ/PCA Dossier	Ch.4 T11, Ch.11.3 T12
Identify Tool (I)	Ch.11.2, Ch.11.3 T3
Installation — Method A (Recommended)	Ch.2.2
Installation — Method B (ZIP)	Ch.2.3
KGE (Kling-Gupta Efficiency)	Ch.5 Alg.4, Ch.7.4, Ch.11.4 Alg.4
Layer Styling (F7)	Ch.11.3 T2
Mekong case study	Ch.13
Model Builder	Ch.12.1
Negotiation AI (GBM)	Ch.4 T8, Ch.11.3 T8, Ch.12.4, Ch.13.2
NSE (Nash-Sutcliffe Efficiency)	Ch.5 Alg.4, Ch.7.4, Ch.11.4 Alg.4
Plugin ID 5040	Cover, Ch.1.1, Ch.10.1, Appendix B, C

Print Layout	Ch.11.8
Processing Toolbox	Ch.3.3, Ch.11.4
Python API (hydrosovereign)	Ch.12.4
Quick Reference Card	Appendix C
SCE-UA Calibration	Ch.5 Alg.4, Ch.11.4 Alg.4, Appendix D
Sentinel-1 GRD	Ch.8, Ch.11.3 T5
Sentinel-2 SR Harmonized	Ch.8, Ch.11.3 T5
SHA-256 integrity chain	Ch.4 T11, Appendix D
SMAP SPL3SMP	Ch.8, Ch.11.3 T5
Toolbar buttons (overview)	Ch.3.1
Troubleshooting	Ch.9, Ch.11.9
UNWC 1997 Articles	Ch.4 T3, Ch.7.1–7.3, Ch.11.3 T3, Ch.13.2
WebGIS Map Generator	Ch.4 T9, Ch.11.3 T9
Workflow Diagram	Appendix D
Xayaburi Dam (Mekong)	Ch.13